

1. Project Title

Smart City E-Governance Network ([e-gov.bd](#))

IP: 192.170.25.0/24

2. Project Objectives

The primary objective of this project was to architect and simulate a production-ready, highly available municipal network. The specific technical engineering goals included:

- **Departmental Segmentation via VLSM:** Utilizing Variable Length Subnet Masking on a single Class C block (192.170.25.0/24) to engineer 17 distinct subnets, logically isolating governmental bodies and securing sensitive operations.
- **Advanced Layer 2 Topologies:** Implementing Multi-VLAN architectures on single physical gateways to enforce zero-trust broadcasting domains across branches with varying clearance levels (e.g., Executive, Defense).
- **Dynamic Routing & Edge Simulation:** Configuring OSPF (Area 0) across all Core and Ministerial gateways for automatic path discovery, alongside a simulated external ISP perimeter featuring a Loopback interface (8.8.8.8/32) for WAN routing validation.
- **Centralized Services Deployment:** Provisioning a localized Data Center farm to handle DNS resolution for the internal e-gov.bd domain, centralized DHCP for dynamic client management, HTTP/HTTPS for citizen portals, and FTP/SMTP for administrative operations.
- **Infrastructure Hardening:** Enforcing strict Layer 2 access controls using Port Security (Sticky MAC addressing) configured with a maximum MAC count of 1 and a violation mode of Shutdown.

3. Introduction

Digital e-governance requires a resilient, secure, and performant underlying network to deliver continuous services to citizens and administrative bodies. This project presents the end-to-end design, implementation, and simulation of a Smart City E-Governance Network using Cisco Packet Tracer.

Built entirely upon the assigned Class C public block (192.170.25.0/24), the infrastructure models a complex ecosystem comprising exactly 17 distinct subnets utilizing IP ranges from .0 to .103. The architecture strictly separates highly classified ministries from standard public sectors utilizing advanced switching techniques, while maintaining seamless interoperability via an OSPF-routed Core Backbone terminating at a simulated external ISP edge.

4. Theoretical Framework and Architecture

To satisfy the project's scalability and security requirements, the implementation followed a structured methodology utilizing standard enterprise networking protocols.

4.1 Variable Length Subnet Masking (VLSM)

To maximize efficiency within the allocated space, VLSM was calculated to eliminate IP waste.

- **Transit Links (/30):** Small /30 point-to-point subnets were deployed exclusively for router-to-router connections across the OSPF backbone to conserve addresses.
- **LAN Segments (/29 & /30):** Variable-sized subnets were allocated for departmental local area networks. For highly restricted endpoints like the President Secretariat, /30 networks were used, while broader departments used /29 segments to tightly bound broadcast traffic.

4.2 Multi-VLAN Switching Topologies

To enforce Layer 2 isolation without excessive hardware overhead, complex VLAN trunking was deployed:

- **Executive Router:** A single physical switch handles two logically separated VLANs. The President Secretariat operates on an isolated /30 network (.28), while the broader PMO, Cabinet, MoPA, and NSI run on a distinct /29 network (.40).
- **Defense Router (Tri-VLAN Architecture):** A high-density switch manages three distinct departmental VLANs. This fully isolates Military/BGB Border Security (.64/29), Law Enforcement/CID (.72/29), and 999 Emergency/Coast Guard Operations (.80/29) from one another.

4.3 Core Routing (OSPF) & Edge Perimeter

OSPF (Open Shortest Path First) Area 0 acts as the internal routing backbone, guaranteeing sub-millisecond route recalculation across all 17 subnets during simulated link failures. The exterior wide-area network (WAN) perimeter is managed by the **ISP-Gateway-Edge** router. This edge gateway routes traffic over the 192.170.25.0/30 transit subnet and features a virtual Loopback interface (8.8.8.8/32) to simulate real-world internet resolution and test exterior packet traversal.

4.4 Centralized Application Services

A dedicated Data Center cluster (192.170.25.32/29) was configured to host essential network services with strict port lockdown:

- **DHCP Server (.34):** Dynamic IP leasing for scalable client deployment across the departments.
- **DNS Server (.35):** Standardized resolution mapping the e-gov.bd domain to the internal server.
- **FTP (.36) & SMTP (.37) Servers:** Secure file operations, configuration backups, and internal mailing.
- **Web Portal (.38):** Hosted the styled internal E-Gov dashboard for simulated citizen services.

5. Implementation and Network IPAM Topology

The physical and logical build within Cisco Packet Tracer adhered strictly to the engineered subnet schema.

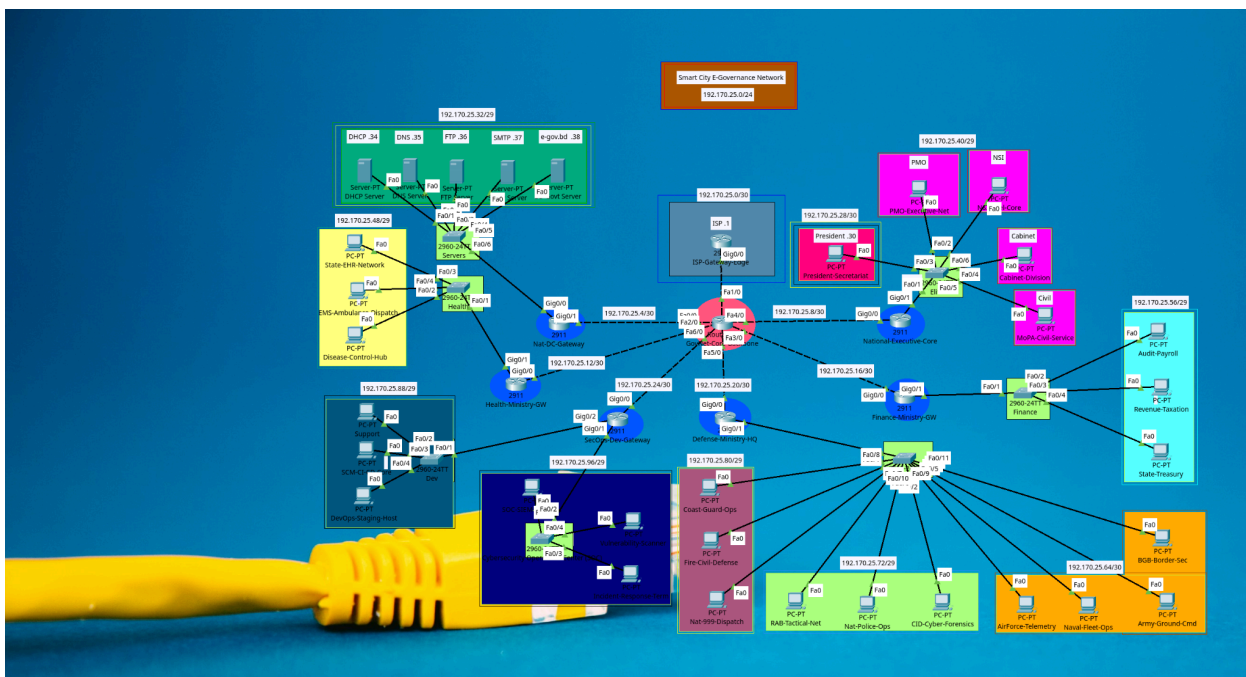


Figure: Smart City E-Governance Network - Realistic Logical Topology.

5.1 Comprehensive IPAM Matrix

The following tables outline the complete addressing allocation strategy executed in the build.

Table 1: Core Backbone Transit Links (/30)

Transit Link Gateway	Subnet IP	IP Range Used	CIDR
ISP Edge Uplink	192.170.25.0	.0 - .3	/30
Server Data Center Gateway	192.170.25.4	.4 - .7	/30
Executive Ministry Gateway	192.170.25.8	.8 - .11	/30
Health Ministry Gateway	192.170.25.12	.12 - .15	/30
Finance Ministry Gateway	192.170.25.16	.16 - .19	/30
Defense Ministry Gateway	192.170.25.20	.20 - .23	/30
SecOps Operations Gateway	192.170.25.24	.24 - .27	/30

Table 2: Departmental Access Networks (/29 & /30)

Department / Logical Zone	Subnet IP	IP Range	CIDR	Key Nodes & Architecture
President Secretariat	192.170.25.28	.28 - .31	/30	Executive VLAN. President IP: .30
Server Farm Core	192.170.25.32	.32 - .39	/29	DHCP(.34), DNS(.35), FTP(.36), SMTP(.37), Web(.38)

PMO & Cabinet Net	192.170.25.40	.40 - .47	/29	Exec VLAN: PM, Cabinet, NSI, MoPA
Health Network	192.170.25.48	.48 - .55	/29	EHR, Ambulance Dispatch, Disease Control
Finance Network	192.170.25.56	.56 - .63	/29	State Treasury, Audit-Payroll, Taxation
Defense (Military)	192.170.25.64	.64 - .71	/29	Tri-VLAN 1: Army, Naval, Air Force, BGB
Defense (Law Enforcement)	192.170.25.72	.72 - .79	/29	Tri-VLAN 2: Police Ops, RAB, CID Forensics
Defense (Emergency)	192.170.25.80	.80 - .87	/29	Tri-VLAN 3: 999 Dispatch, Coast Guard
SecOps (Dev Switch)	192.170.25.88	.88 - .95	/29	CI/CD Core, DevOps Staging Host
SecOps (Cyber Switch)	192.170.25.96	.96 - .103	/29	SOC SIEM, Vulnerability Scanners, IR

6. Testing, Validation, and Troubleshooting

Network validation verified physical link states, OSPF convergence, and end-to-end application reachability for all implemented Data Center services.

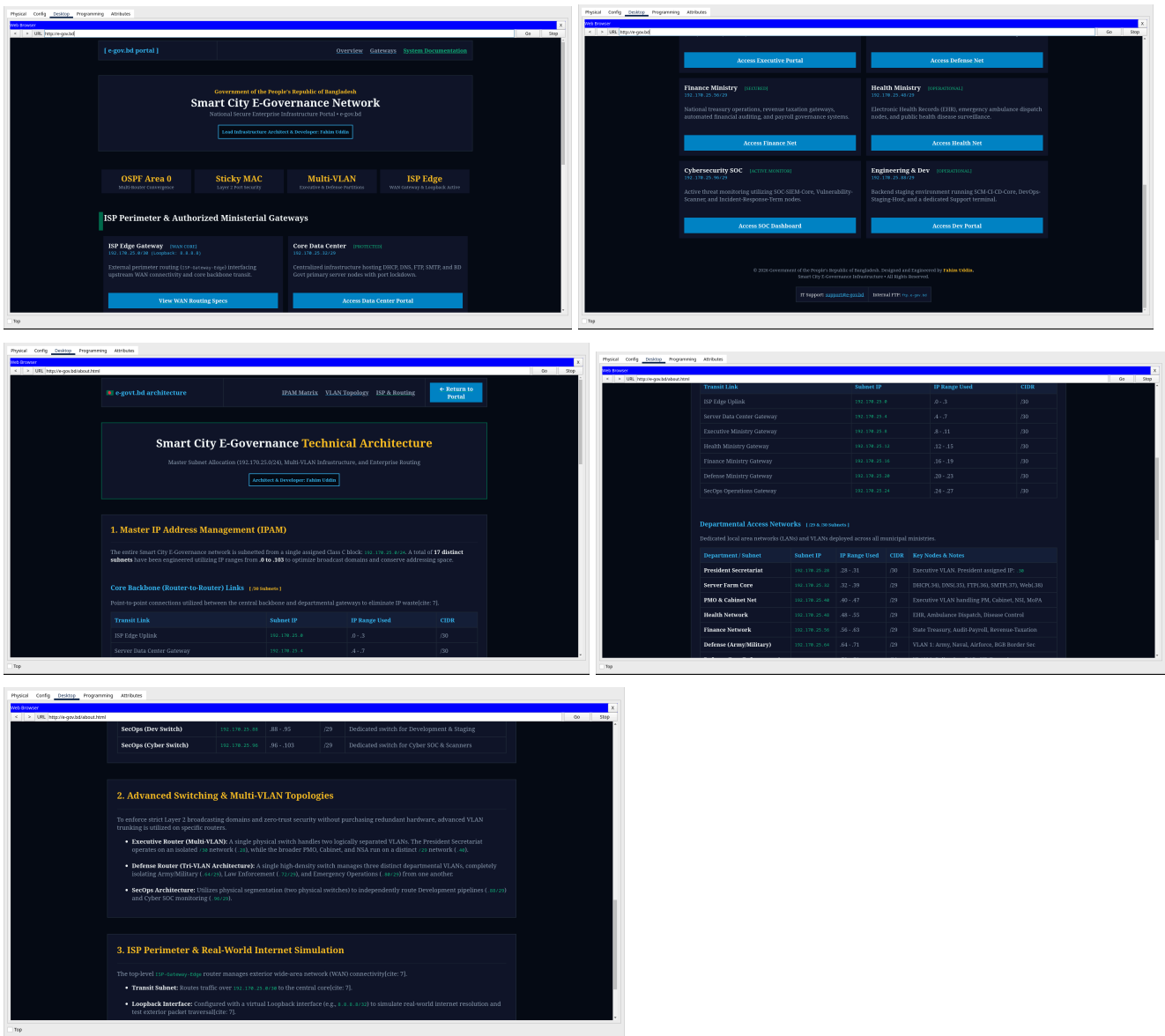
6.1 Overcoming Simulation Limitations (Buffer Full Error)

During testing, Cisco Packet Tracer triggered a "Buffer Full" constraint. This occurred because the high-density backbone continuously generated OSPF and CDP link-state updates in the background, exhausting the simulation memory. The resolution required clicking "Clear Event List" and applying strict protocol filters within the Simulation Panel to hide background routing traffic. By enabling only specific application-layer protocols (ICMP, HTTP, FTP, SMTP) visibility, successful packet tracing of user requests across the routed ministerial gateways was achieved without crashing the interface.

6.2 Validation Evidence

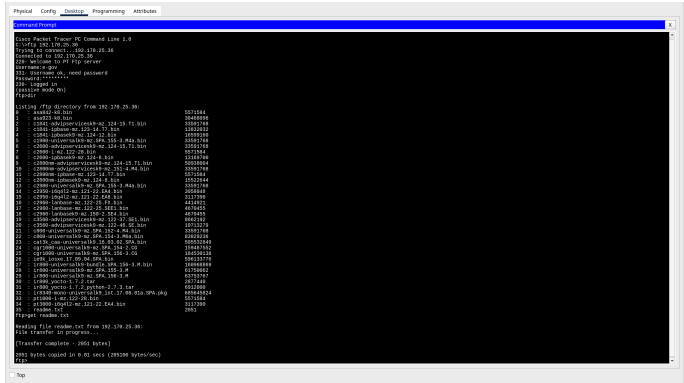
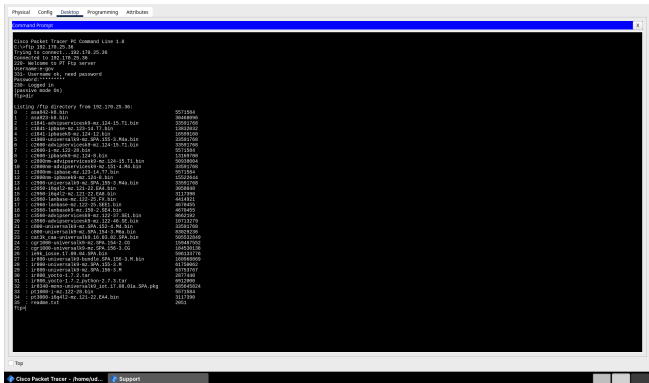
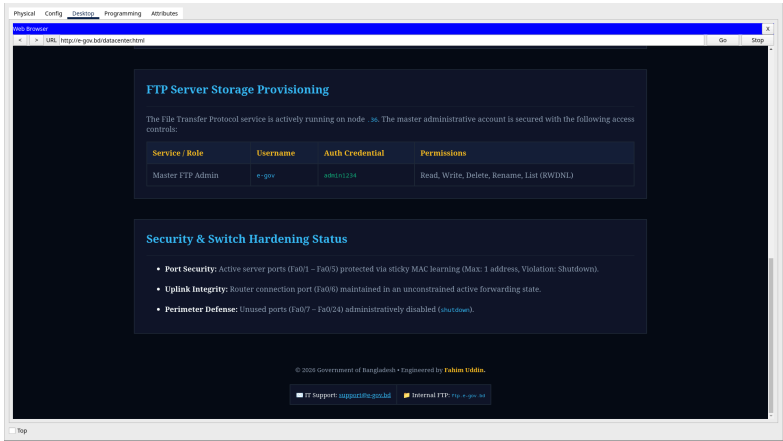
A. DNS and HTTP Portal Resolution: Simulated client machines successfully queried the localized DNS server and retrieved the hosted E-Governance portal. The custom HTML interface, featuring the top navigation banner and official government headers, rendered correctly under the [\[http://e-gov.bd\]](http://e-gov.bd)(<http://e-gov.bd>) domain name. The lower segment of the portal successfully populated the interactive gateway access grid, confirming the centralized web server's full CSS styling and HTML parsing capabilities.

Reference Visuals:



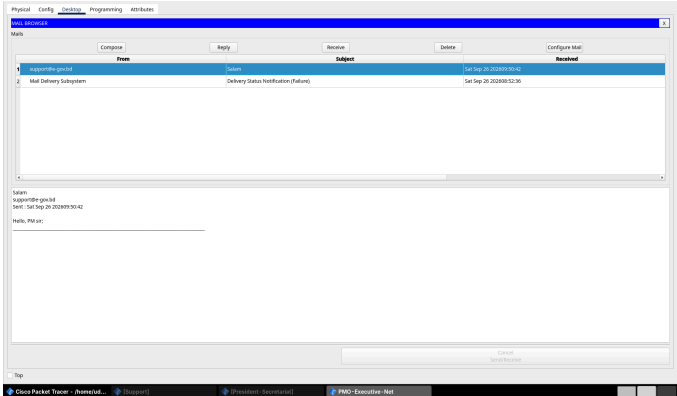
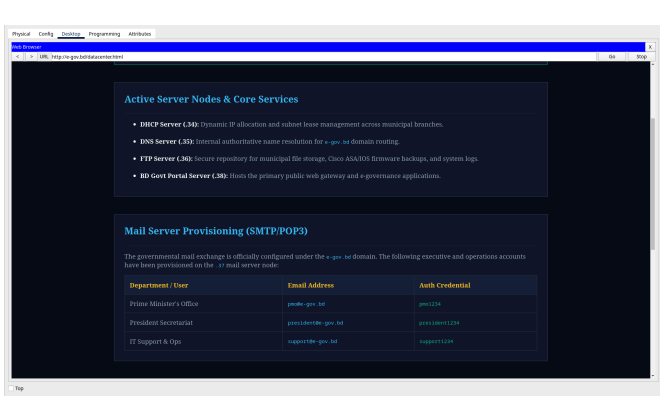
B. FTP Server Connectivity and Directory Access: File Transfer Protocol connectivity was validated via the command prompt of a client PC. The system successfully executed an active login to the FTP server at 192.170.25.36 using the configured e-gov credentials. After logging in, the client retrieved the full **directory** listing, confirming read access to stored configuration binaries (e.g., [asa842-k8.bin](#), [readme.txt](#)) across the routed network.

Reference Visual: readme.txt



C. SMTP Mail Routing and Delivery: Internal email routing was verified using the Packet Tracer Mail Browser client. The mail service successfully processed an inbound message from support@e-gov.bd with the subject line "Salam" and the body addressing "PM sir". The presence of a delivery status notification in the inbox confirms that the SMTP server (.37) is actively handling mail queues and bounce responses.

Reference Visual:



D. External Loopback & Edge Reachability

Validation of transit routing to the simulated WAN perimeter.

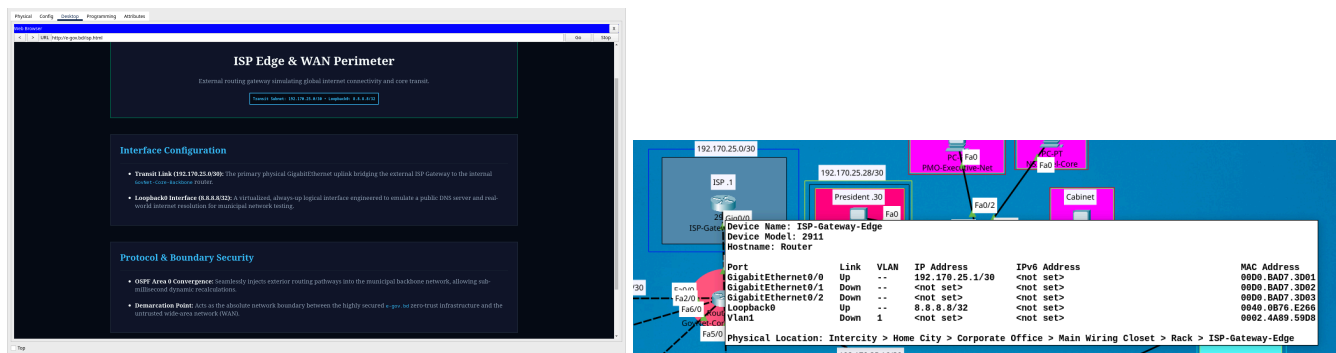
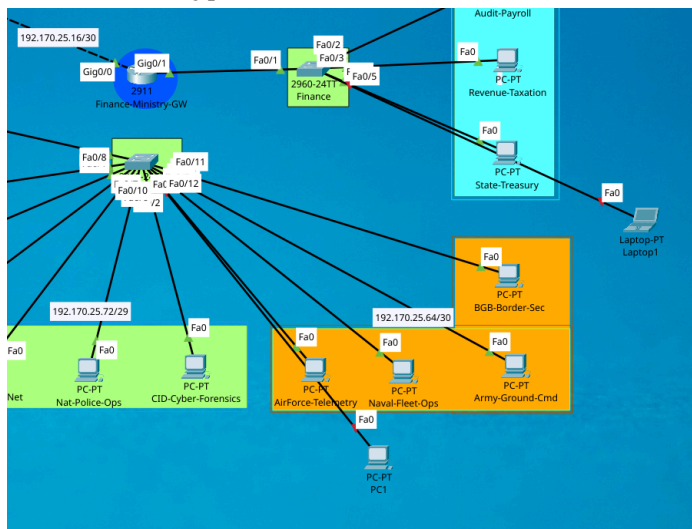


Figure: ICMP validation confirming WAN edge connectivity to the 8.8.8.8/32 loopback interface.

E. Layer 2 Port Security

Administrative hardening was verified on the access layer. Port Security is enabled on the production ministerial access switches, locking the port to the 'sticky' MAC address of the assigned client PC with a maximum count of 1 and a violation type of Shutdown.



7. Conclusion

This exhaustive simulation of a 17-subnet architecture proves the technical viability of the design. The meticulous execution of VLSM ensured zero IP waste within the allocated /24 block, while the deployment of multi-VLAN switching isolated critical intelligence and defense branches. OSPF provided robust automated pathing across the core transit links, allowing the highly secured data center to deliver essential application-layer services (HTTP, DNS, DHCP, FTP, SMTP) seamlessly to all departmental endpoints. By layering dynamic routing with strict MAC-level access security (max count 1/shutdown) and verifying external reachability via Loopback WAN testing, the implemented architecture successfully provides a highly resilient, enterprise-grade foundation for municipal e-governance operations.

8. References

- Cisco Press. (n.d.). *Cisco Packet Tracer*. Cisco Networking Academy.
- Cisco Press. (2019). *CCNA 200-301 Official Cert Guide Library*. OSPF Area Design and Configuration.
- Doyle, J. (2006). *Routing TCP/IP, Volume I* (2nd ed.). Cisco Press. Link-State Dynamic Routing Fundamentals.